ExamGOAL
PYQ PDF

Free · With answer key

JEE MAIN · CHEMISTRY · CHAPTER-WISE PYQS

Salt Analysis

JEE Main Chemistry · Previous Year Questions with Answer Key

52	47	2002 – 2026	1	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

MCQ (Single Correct Answer)	48 questions
Numerical	3 questions
Out of syllabus	1 question
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

MCQ (Single Correct Answer) 48 questions

Q1. JEE Main 2026 (Online) 4th April Evening Shift

Among $Fe^{3+}, Pb^{2+}, Cu^{2+}$ and Mn^{2+}, identify the one that gets precipitated out while passing H_2S in presence of NH_4OH as group reagent. The highest possible oxidation state of the corresponding metal is

Open solution

(A) +3 (B) +4 (C) +2 (D) +7

Q2. JEE Main 2026 (Online) 2nd April Morning Shift

A salt with few drops of conc. HCl gives apple green colour in flame test. The group precipitate of the salt is dissolved in acetic acid and treated with K_2CrO_4 to give yellow precipitate. When the sodium carbonate extract of the salt solution is heated with conc. HNO_3 and ammonium molybdate, it resulted a canary yellow precipitate. The cation and anion present in the salt are respectively,

Open solution

(A) Ca^{2+} and SO_4^{2-} (B) Ba^{2+} and PO_4^{3-} (C) Mn^{2+} and PO_4^{3-} (D) Ba^{2+} and SO_4^{2-}

Q3. JEE Main 2026 (Online) 28th January Evening Shift

Consider the following reactions

$$Na_2B_4O_7 \xrightarrow{\Delta} 2X + Y$$

$$CuSO_4 + Y \xrightarrow{\text{Non-Luminous flame}} Z + SO_3$$

$$2Z + 2X + \text{Carbon} \xrightarrow{\text{Luminous flame}} 2Q + Na_2B_4O_7 + CO$$

The oxidation states of Cu in Z and Q, respectively are :

Open solution

(A) +2 and +1 (B) +1 and +1 (C) +1 and +2 (D) +2 and +2

Q4. JEE Main 2026 (Online) 28th January Morning Shift

Given below are two statements:

Statement I : Griess-Ilosvay test is used for the detection of nitrite ion, which involves the use of sulphanilic acid and α-naphthylamine reagent.

Statement II : In the above test, sulphanilic acid is diazotized by the acidified nitrite ion, which on further coupling with α-naphthylamine forms an azo-dye.

In the light of the above statements, choose the correct answer from the options given below

Open solution

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are true

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q5. JEE Main 2026 (Online) 24th January Evening Shift

In the Group analysis of cations, Ba^{2+} & Ca^{2+} are precipitated respectively as

(A) sulphide & sulphide

(B) hydroxide & carbonate

(C) chromate & sulphide

(D) carbonate & carbonate

Open solution

Q6. JEE Main 2026 (Online) 24th January Morning Shift

Consider three metal chlorides x, y and z , where x is water soluble at room temperature, y is sparingly soluble in water at room temperature and z is soluble in hot water. x, y and z are respectively

(A) $AgCl, Hg_2Cl_2$ and $PbCl_2$

(B) $CuCl_2, AgCl$ and $PbCl_2$

(C) $AlCl_3, PbCl_2$ and $BaCl_2$

(D) $MgCl_2, AgCl$ and $AlCl_3$

Open solution

Q7. JEE Main 2025 (Online) 8th April Evening Shift

Match the **LIST-I** with **LIST-II**

LIST-I (Reagent)	LIST-II (Functional Group detected)
A. Sodium bicarbonate solution	I. double bond/unsaturation
B. Neutral ferric chloride	II. carboxylic acid
C. ceric ammonium nitrate	III. phenolic - OH
D. alkaline $KMnO_4$	IV. alcoholic - OH

Choose the *correct* answer from the options given below:

(A) A-II, B-III, C-I, D-IV

(B) A-II, B-IV, C-III, D-I

(C) A-III, B-II, C-IV, D-I

(D) A-II, B-III, C-IV, D-I

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q8. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

Which of the following compounds is least likely to give effervescence of CO_2 in presence of aq. $NaHCO_3$?

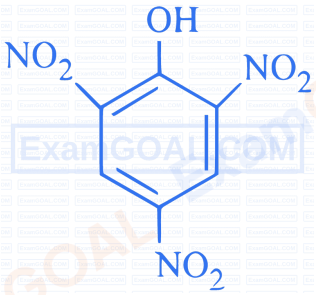
(A)



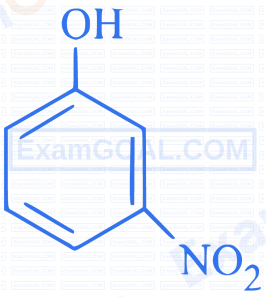
(B)



(C)



(D)



Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q9. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

Given below are two statements :

Statement I : Mohr's salt is composed of only three types of ions-ferrous, ammonium and sulfate.

Statement II : If the molar conductance at infinite dilution for ferrous, ammonium and sulfate ions are x_1, x_2 and x_3 $\mathrm{S\ cm^2\ mol^{-1}}$, respectively then the molar conductance for Mohr's salt solution at infinite dilution would be given by $x_1 + x_2 + 2x_3$

In the light of the given statements, choose the correct answer from the options given below :

(A) Both Statements I and Statement II are true

(B) Statement I is true but Statement II are false

(C) Both Statements I and Statement II are false

(D) Statement I is false but Statement II is true

Q10. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

When a salt is treated with sodium hydroxide solution it gives gas X . On passing gas X through reagent Y a brown coloured precipitate is formed. X and Y respectively, are

(A) $\mathrm{X = NH_3}$ and $\mathrm{Y = HgO}$

(B) $\mathrm{X = NH_4Cl}$ and $\mathrm{Y = KOH}$

(C) $\mathrm{X = NH_3}$ and $\mathrm{Y = K_2HgI_4 + KOH}$

(D) $\mathrm{X = HCl}$ and $\mathrm{Y = NH_4Cl}$

Q11. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

Compounds that should not be used as primary standards in titrimetric analysis are :

A. $\mathrm{Na_2Cr_2O_7}$

B. Oxalic acid

C. NaOH

D. $\mathrm{FeSO_4 \cdot 6H_2O}$

E. Sodium tetraborate

Choose the most appropriate answer from the options given below:

(A) C, D and E Only (B) B and D Only (C) D and E Only (D) A, C and D Only

Q12. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Formation of $\mathrm{Na_4\,[Fe(CN)_5NOS]}$, a purple coloured complex formed by addition of sodium nitroprusside in sodium carbonate extract of salt indicates the presence of :

(A) Sulphide ion (B) Sulphite ion (C) Sulphate ion (D) Sodium ion

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q13. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

Choose the correct tests with respective observations.

(A) $CuSO_4$ (acidified with acetic acid) $+K_4[Fe(CN)_6] \rightarrow$ Chocolate brown precipitate.

(B) $FeCl_3 + K_4[Fe(CN)_6] \rightarrow$ Prussian blue precipitate.

(C) $ZnCl_2 + K_4[Fe(CN)_6]$, neutralised with $NH_4OH \rightarrow$ White or bluish white precipitate.

(D) $MgCl_2 + K_4[Fe(CN)_6] \rightarrow$ Blue precipitate.

(E) $BaCl_2 + K_4[Fe(CN)_6]$, neutralised with $NaOH \rightarrow$ White precipitate.

Choose the correct answer from the options given below :

(A) A, D and E only **(B)** A, B and C only **(C)** B, D and E only **(D)** C, D and E only

Q14. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

Identify the inorganic sulphides that are yellow in colour :

(A) $(NH_4)_2S$

(B) PbS

(C) CuS

(D) As_2S_3

(E) As_2S_5

Choose the **correct** answer from the options given below :

(A) (D) and (E) only **(B)** (A) and (B) only

(C) (A), (D) and (E) only **(D)** (A) and (C) only

Q15. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

Find the compound 'A' from the following reaction sequences.

$$A \xrightarrow{\text{aqua-regia}} B \xrightarrow[(2)\ AcOH]{(1)\ KNO_2|NH_4OH} \text{yellow ppt}$$

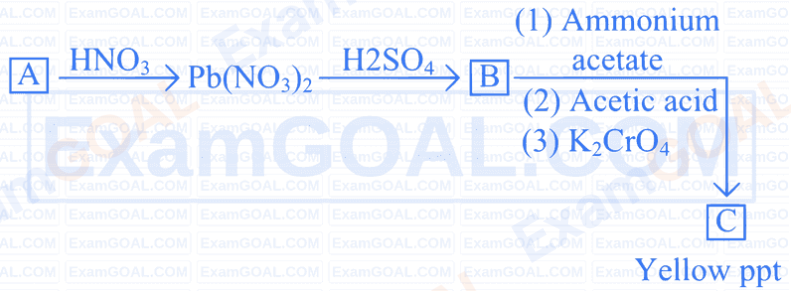
(A) NiS **(B)** MnS **(C)** ZnS **(D)** CoS

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q16. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

Identify A, B and C in the given below reaction sequence



(A) $PbCl_2, PbSO_4, PbCrO_4$

(B) $PbS, PbSO_4, Pb(CH_3COO)_2$

(C) $PbS, PbSO_4, PbCrO_4$

(D) $PbCl_2, Pb(SO_4)_2, PbCrO_4$

Q17. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

Given below are two statements :

Statement (I) : Nitrogen, sulphur, halogen and phosphorus present in an organic compound are detected by Lassaigne's Test.
Statement (II) : The elements present in the compound are converted from covalent form into ionic form by fusing the compound with Magnesium in Lassaigne's test.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are true

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q18. JEE Main 2024 (Online) 8th April Morning Shift

Open solution

Match List I with List II

	LIST I (Name of the test)		LIST II (Reaction sequence involved) [M is metal]
A.	Borax bead test	I.	$MCO_3 \rightarrow MO \xrightarrow[+\Delta]{Co(NO_3)_2} CoO \cdot MO$
B.	Charcoal cavity test	II.	$MCO_3 \rightarrow MCl_2 \rightarrow M^{2+}$
C.	Cobalt nitrate test	III.	$MSO_4 \xrightarrow[\Delta]{Na_2B_4O_7} M(BO_2)_2 \rightarrow MBO_2 \rightarrow M$
D.	Flame test	IV.	$MSO_4 \xrightarrow[\Delta]{Na_2CO_3} MCO_3 \rightarrow MO \rightarrow M$

Choose the correct answer from the options given below:

(A) A-III, B-I, C-IV, D-II

(B) A-III, B-IV, C-I, D-II

(C) A-III, B-II, C-IV, D-I

(D) A-III, B-I, C-II, D-IV

Q19. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

During the detection of acidic radical present in a salt, a student gets a pale yellow precipitate soluble with difficulty in NH_4OH solution when sodium carbonate extract was first acidified with dil. HNO_3 and then $AgNO_3$ solution was added. This indicates presence of :

(A) I^-

(B) CO_3^{2-}

(C) Cl^-

(D) Br^-

Q20. JEE Main 2024 (Online) 6th April Morning Shift

Open solution

Match List I with List II

	LIST I (Precipitating reagent and conditions)		LIST II (Cation)
A.	$NH_4Cl + NH_4OH$	I.	Mn^{2+}
B.	$NH_4OH + Na_2CO_3$	II.	Pb^{2+}
C.	$NH_4OH + NH_4Cl + H_2S$ gas	III.	Al^{3+}
D.	dilute HCl	IV.	Sr^{2+}

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II

(B) A-III, B-IV, C-II, D-I

(C) A-IV, B-III, C-II, D-I

(D) A-IV, B-III, C-I, D-II

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q21. JEE Main 2024 (Online) 4th April Morning Shift

In the precipitation of the iron group (III) in qualitative analysis, ammonium chloride is added before adding ammonium hydroxide to :

(A) decrease concentration of ^{-}OH ions

(B) increase concentration of Cl^{-} ions

(C) prevent interference by phosphate ions

(D) increase concentration of NH_4^{+} ions

Open solution

Q22. JEE Main 2024 (Online) 31st January Morning Shift

The compound that is white in color is

(A) ammonium sulphide

(B) ammonium arsinomolybdate

(C) lead iodide

(D) lead sulphate

Open solution

Q23. JEE Main 2024 (Online) 30th January Morning Shift

Given below are two statements :

Statement (I) : The gas liberated on warming a salt with dil H_2SO_4, turns a piece of paper dipped in lead acetate into black, it is a confirmatory test for sulphide ion.

Statement (II) : In statement-I the colour of paper turns black because of formation of lead sulphite.
In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are true

(D) Statement I is false but Statement II is true

Open solution

Q24. JEE Main 2023 (Online) 13th April Evening Shift

In the wet tests for detection of various cations by precipitation, Ba^{2+} cations are detected by obtaining precipitate of

(A) $BaSO_4$

(B) $Ba(OAc)_2$

(C) $BaCO_3$

(D) $Ba(ox)$: Barium oxalate

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q25. JEE Main 2023 (Online) 11th April Morning Shift

When a solution of mixture having two inorganic salts was treated with freshly prepared ferrous sulphate in acidic medium, a dark brown ring was formed whereas on treatment with neutral $FeCl_3$, it gave deep red colour which disappeared on boiling and a brown red ppt was formed. The mixture contains :

(A) SO_3^{2-} & $C_2O_4^{2-}$
(B) CH_3COO^- & NO_3^-
(C) $C_2O_4{}^{2-}$ & NO_3^-
(D) SO_3^{2-} & CH_3COO^-

Open solution

Q26. JEE Main 2023 (Online) 6th April Evening Shift

Formation of which complex, among the following, is not a confirmatory test of Pb^{2+} ions :

(A) lead chromate (B) lead iodide (C) lead sulphate (D) lead nitrate

Open solution

Q27. JEE Main 2023 (Online) 31st January Evening Shift

Given below are two statements :

Statement I: Upon heating a borax bead dipped in cupric sulphate in a luminous flame, the colour of the bead becomes green

Statement II: The green colour observed is due to the formation of copper(I) metaborate

In light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are false
(B) Both Statement I and Statement II are true
(C) Statement I is false but Statement II is true
(D) Statement I is true but Statement II is false

Open solution

Q28. JEE Main 2023 (Online) 30th January Morning Shift

In the wet tests for identification of various cations by precipitation, which transition element cation doesn't belong to group IV in qualitative inorganic analysis?

(A) Co^{2+} (B) Ni^{2+} (C) Zn^{2+} (D) Fe^{3+}

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q29. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

Match List I with List II

	List I Elements		List II Colour imparted to the flame
A.	K	I.	Brick Red
B.	Ca	II.	Violet
C.	Sr	III.	Apple Green
D.	Ba	IV.	Crimson Red

Choose the correct answer from the options given below :

(A) A-II, B-IV, C-I, D-III

(B) A-II, B-I, C-IV, D-III

(C) A-II, B-I, C-III, D-IV

(D) A-IV, B-III, C-II, D-I

Q30. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

Match the List-I with List-II :

	List-I Cations		List-II Group reagents
A.	Pb^{2+}, Cu^{2+}	i)	H_2S gas in presence of dilute HCl
B.	Al^{3+}, Fe^{3+}	ii)	$(NH_4)_2CO_3$ in presence of NH_4OH
C.	Co^{2+}, Ni^{2+}	iii)	NH_4OH in presence of NH_4Cl
D.	Ba^{2+}, Ca^{2+}	iv)	H_2S in presence of NH_4OH

Correct match is -

(A) A $\rightarrow$ i, B $\rightarrow$ iii, C $\rightarrow$ iv, D $\rightarrow$ ii

(B) A $\rightarrow$ i, B $\rightarrow$ iii, C $\rightarrow$ ii, D $\rightarrow$ iv

(C) A $\rightarrow$ iii, B $\rightarrow$ i, C $\rightarrow$ iv, D $\rightarrow$ ii

(D) A $\rightarrow$ iv, B $\rightarrow$ ii, C $\rightarrow$ iii, D $\rightarrow$ i

Q31. JEE Main 2022 (Online) 29th June Evening Shift

Open solution

A white precipitate was formed when $BaCl_2$ was added to water extract of an inorganic salt. Further, a gas 'X' with characteristic odour was released when the formed white precipitate was dissolved in dilute HCl. The anion present in the inorganic salt is

(A) I^- **(B)** SO_3^{2-} **(C)** S^{2-} **(D)** NO_2^-

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q32. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

Match List I with List II.

	List-I (Anion)		List-II (gas evolved on reaction with dil.H_2SO_2)
(A)	CO_3^{2-}	I.	Colourless gas which turns lead acetate paper black.
(B)	S^{2-}	II.	Colourless gas which turns acidified potassium dichromate solution green.
(C)	SO_3^{2-}	III.	Brown fumes which turns acidified KI solution containing starch blue.
(D)	NO_2^-	IV.	Colourless gas evolved with brisk effervescence, which turns lime water milky.

Choose the correct answer from the options given below :

(A) A-III, B-I, C-II, D-IV

(B) A-II, B-I, C-IV, D-III

(C) A-IV, B-I, C-III, D-II

(D) A-IV, B-I, C-II, D-III

Q33. JEE Main 2022 (Online) 24th June Evening Shift

Open solution

In the flame test of a mixture of salts, a green flame with blue centre was observed. Which one of the following cations may be present?

(A) Cu^{2+} **(B)** Sr^{2+} **(C)** Ba^{2+} **(D)** Ca^{2+}

Q34. JEE Main 2022 (Online) 24th June Morning Shift

Open solution

During the qualitative analysis of salt with cation y^{2+}, addition of a reagent (X) to alkaline solution of the salt gives a bright red precipitate. The reagent (X) and the cation (y^{2+}) present respectively are :

(A) Dimethylglyoxime and Ni^{2+}

(B) Dimethylglyoxime and Co^{2+}

(C) Nessler's reagent and Hg^{2+}

(D) Nessler's reagent and Ni^{2+}

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q35. JEE Main 2021 (Online) 31st August Evening Shift

Open solution

Match List - I with List - II :

	List - I (Metal Ion)		List - II (Group of Qualitative analysis)
(a)	Mn^{2+}	(i)	Group - III
(b)	As^{3+}	(ii)	Group - IIA
(c)	Cu^{2+}	(iii)	Group - IV
(d)	Al^{3+}	(iv)	Group - IIB

Choose the most appropriate answer from the options given below :

(A) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(B) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(C) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

(D) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

Q36. JEE Main 2021 (Online) 27th July Evening Shift

Open solution

To an aqueous solution containing ions such as Al^{3+}, Zn^{2+}, Ca^{2+}, Fe^{3+}, Ni^{2+}, Ba^{2+} and Cu^{2+} was added conc. HCl, followed by H_2S.

The total number of cations precipitated during this reaction is/are :

(A) 1 **(B)** 3 **(C)** 4 **(D)** 2

Q37. JEE Main 2021 (Online) 20th July Evening Shift

Open solution

Cu^{2+} salt reacts with potassium iodide to give

(A) Cu_2I_2 **(B)** Cu_2I_3 **(C)** CuI **(D)** $Cu(I_3)_2$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q38. JEE Main 2021 (Online) 25th February Evening Shift

Open solution

Given below are two statements :

Statement I : The identification of Ni^{2+} is carried out by dimethyl glyoxime in the presence of NH_4OH.

Statement II : The dimethyl glyoxime is a bidentate neutral ligand.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true.

(B) Statement I is true but Statement II is false.

(C) Both Statement I and Statement II are false.

(D) Both Statement I and Statement II are true.

Q39. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

Match the following :

Test/Method	Reagent
(i) Lucas Test	(a) $C_6H_5SO_2Cl$/ aq. KOH
(ii) Dumas method	(b) HNO_3/ $AgNO_3$
(iii) Kjeldahl's method	(c) CuO/CO_2
(iv) Hinsberg test	(d) Conc. HCl and $ZnCl_2$
	(e) H_2SO_4

(A) (i)-(b), (ii)-(d), (iii)-(e), (iv)-(a)

(B) (i)-(d), (ii)-(c), (iii)-(e), (iv)-(a)

(C) (i)-(b), (ii)-(a), (iii)-(c), (iv)-(d)

(D) (i)-(d), (ii)-(c), (iii)-(b), (iv)-(e)

Q40. JEE Main 2020 (Online) 7th January Morning Slot

Open solution

A solution of m-chloroaniline, m-chlorophenol and m-chlorobenzoic acid in ethyl acetate was extracted initially with a saturated solution of $NaHCO_3$ to give fraction A. The left over organic phase was extracted with dilute NaOH solution to give fraction B. The final organic layer was labelled as fraction C. Fractions A,B and C, contain respectively :

(A) m-chlorobenzoic acid, m-chloroaniline and m-chlorophenol.

(B) m-chlorobenzoic acid, m-chlorophenol and m-chloroaniline.

(C) m-chlorophenol, m-chlorobenzoic acid and m-chloroaniline.

(D) m-chloroaniline, m-chlorobenzoic acid and m-chlorophenol.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q41. JEE Main 2017 (Online) 8th April Morning Slot

A solution containing a group-IV cation gives a precipitate on passing H_2S. A solution of this precipitate in dil.HCl produces a white precipitate with NaOH solution and bluish-white precipitate with basic potassium ferrocyanide. The cation is :

(A) Co^{2+} (B) Ni^{2+} (C) Mn^{2+} (D) Zn^{2+}

Open solution

Q42. JEE Main 2015 (Online) 11th April Morning Slot

A pink coloured salt turns blue on heating. The presence of which cation is most likely ?

(A) Cu^{2+} (B) Fe^{2+} (C) Zn^{2+} (D) Co^{2+}

Open solution

Q43. JEE Main 2015 (Online) 10th April Morning Slot

An aqueous solution of a salt X turns blood red on treatment with SCN^- and blue on treatment with $K_4[Fe(CN)_6]$. X also gives a positive chromyl chloride test. The salt X is :

(A) $CuCl_2$ (B) $FeCl_3$ (C) $Cu(NO_3)_2$ (D) $Fe(NO_3)_3$

Open solution

Q44. JEE Main 2015 (Online) 10th April Morning Slot

The cation that will not be precipitated by H_2S in the presence of dil HCl is :

(A) Cu^{2+} (B) Pb^{2+} (C) As^{3+} (D) Co^{2+}

Open solution

Q45. JEE Main 2013 (Online) 22th April Morning Slot

Which of the following statements is incorrect?

(A) Fe^{2+} ion also gives blood red colour with SCN^- ion.

(B) Fe^{3+} ion also gives blood red colour with SCN^- ion.

(C) On passing H_2S into Na_2ZnO_2 solution a white ppt of ZnS is formed.

(D) Cupric ion reacts with excess of ammonia solution to give deep blue colour of $[Cu(NH_3)_4]^{2+}$ ion.

Open solution

Q46. JEE Main 2013 (Online) 9th April Morning Slot

Sodium Carbonate cannot be used in place of $(NH_4)_2CO_3$ for the identification of Ca^{2+}, Ba^{2+} and Sr^{2+} ions (in group V) during mixture analysis because :

(A) Mg^{2+} ions will also be precipitated.

(B) Concentration of CO_3^{2-} ions is very low.

(C) Sodium ions will react with acid radicals.

(D) Na^+ ions will interfere with the detection of Ca^{2+}, Ba^{2+}, Sr^{2+} ions.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q47. JEE Main 2013 (Online) 9th April Morning Slot

Potassium dichromate when heated with concentrated sulphuric acid and a soluble chloride, gives brown-red vapours of :

(A) CrO_3 (B) $CrCl_3$ (C) CrO_2Cl_2 (D) Cr_2O_3

Open solution

Q48. AIEEE 2002

How do we differentiate between Fe^{3+} and Cr^{3+} in group III?

(A) by taking excess of NH_4OH solution

(B) by increasing NH_4^+ ion concentration

(C) by decreasing OH^- ion concentration

(D) both (b) and (c)

Open solution

Numerical 3 questions

Q49. JEE Main 2025 (Online) 24th January Morning Shift

Among the following cations, the number of cations which will give characteristic precipitate in their identification tests with $K_4[Fe(CN)_6]$ is _____.

$Cu^{2+}, Fe^{3+}, Ba^{2+}, Ca^{2+}, NH_4^+, Mg^{2+}, Zn^{2+}$

Open solution

Q50. JEE Main 2024 (Online) 1st February Morning Shift

The number of white coloured salts, among the following is

(a) $SrSO_4$

(b) $Mg(NH_4)PO_4$

(c) $BaCrO_4$

(d) $Mn(OH)_2$

(e) $PbSO_4$

(f) $PbCrO_4$

(g) $AgBr$

(h) PbI_2

(i) CaC_2O_4

(j) $[Fe(OH)_2(CH_3COO)]$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q51. JEE Main 2024 (Online) 30th January Evening Shift

Number of metal ions characterized by flame test among the following is _____.

$Sr^{2+}, Ba^{2+}, Ca^{2+}, Cu^{2+}, Zn^{2+}, Co^{2+}, Fe^{2+}$

Open solution

Out of syllabus 1 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q52. JEE Main 2018 (Online) 15th April Morning Slot · **MCQ (Single Correct Answer)**

A white sodium salt dissolves readily in water to give a solution which is neutral to litmus. When silver nitrate solution is added to the aformentioned solution, a white precipitate is obtained which does not dissolve in dil. nitric acid. The anion is :

Open solution

(A) CO_3^{2-} (B) SO_4^{2-} (C) Cl^- (D) S^{2-}

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

MCQ (Single Correct Answer)

1. **D**	2. **B**	3. **A**	4. **D**	5. **D**
6. **B**	7. **D**	8. **D**	9. **B**	10. **C**
11. **D**	12. **A**	13. **B**	14. **A**	15. **D**
16. **C**	17. **B**	18. **B**	19. **D**	20. **A**
21. **A**	22. **D**	23. **B**	24. **C**	25. **B**
26. **D**	27. **A**	28. **D**	29. **B**	30. **A**
31. **B**	32. **D**	33. **A**	34. **A**	35. **B**
36. **A**	37. **C**	38. **D**	39. **B**	40. **B**
41. **D**	42. **D**	43. **B**	44. **D**	45. **A**
46. **A**	47. **C**	48. **B**		

Numerical

49. **4**	50. **5**	51. **4**

Out of syllabus

52. **C**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs